

TASK 3 – SECURITY & MONITORING

Terraform-Managed Azure Subscription Baseline

1. Objective

To introduce identity-based access control and monitoring capabilities into the Azure baseline. This task covers Week 5 and Week 6 and demonstrates managed identity, Azure RBAC, Log Analytics and VM diagnostic metrics.

2. Week 5 – Security & Access Management

A user-assigned managed identity named **id-terraform-baseline** was provisioned and assigned the Azure **Reader** role at the resource-group scope. The identity provides a separate Azure identity that can be used by workloads without embedding credentials in application code. Terraform manages both the identity and its role assignment.

3. Week 6 – Monitoring & Logging

A Log Analytics Workspace named **law-terraform-baseline** was created in Southeast Asia with the PerGB2018 SKU and 30-day retention. A diagnostic setting named **vm-diagnostics** sends the VM's **AllMetrics** category to the Log Analytics Workspace. The VM exposes metrics through Azure monitoring; the current implementation does not configure VM log categories because the deployed VM exposed AllMetrics as the available diagnostic category.

4. Security and Monitoring Flow

Layer	Resource	Purpose
Identity	id-terraform-baseline	Provides a user-assigned Azure identity
Access	Reader role	Grants read access at resource-group scope
Monitoring	law-terraform-baseline	Central Log Analytics workspace
Diagnostics	vm-diagnostics	Routes VM AllMetrics to the workspace

5. Security Design

The project separates identity from infrastructure resources by using a user-assigned managed identity and an explicit RBAC assignment. This avoids placing credentials directly in the Terraform configuration for the workload identity. The current Reader permission is intentionally limited to read access at the resource-group scope.

6. Monitoring Design

The Log Analytics Workspace provides a central monitoring destination, while the diagnostic setting connects the VM to that workspace for AllMetrics. This provides a foundation for observing resource health and can be extended with additional monitoring rules, alerts and log collection in a production design.

7. Expected Result

The Azure baseline contains a managed identity with RBAC access and a monitoring workspace connected to the VM through diagnostic settings. Both security and monitoring resources are managed through Terraform.

8. Evidence

The accompanying README contains the recommended screenshot checklist. Do not include secrets, tokens or private keys in screenshots.